

Does Financial News Sentiment Improve Neural-SDE Option Pricing?

Evidence from a Corrected Two-Stage Calibration Pipeline

Ethan Itkis Matthew Lascoe Stephen Linehan-Reckford Om Patel
Samuel Shlyam

Stevens Institute of Technology

August 24, 2026

Abstract

We test whether daily financial-news sentiment, extracted from Refinitiv/LSEG news via FinBERT, improves the pricing accuracy of neural stochastic-volatility models calibrated to S&P 500 index options. Following the two-stage deep-calibration framework of [Horvath et al. \[2021\]](#), we pre-train a convolutional network offline on synthetic implied-volatility surfaces generated from four stochastic-volatility families—Heston, Bates, one-factor Bergomi and rough Bergomi—and then fine-tune it online against historical SPX surfaces through a fully differentiable pricer. Two arms are compared under identical architecture and hyperparameters: a baseline that sees only the implied-volatility surface, and a sentiment arm that additionally receives a daily sentiment channel. We evaluate 32 (model, regime, arm) cells across four market regimes spanning 2010–2022.

Offline, the sentiment channel is genuinely informative: the sentiment arm attains 4–25% lower validation loss than the baseline across all four model families. Online, the picture is far more equivocal. In-sample, Diebold-Mariano tests find statistically significant differences in 11 of 16 cells, but split 7 favourable and 4 unfavourable with effect sizes below 5%. Under expanding-window walk-forward evaluation—the more demanding test—7 cells favour sentiment and 6 favour the baseline, and the in-sample and out-of-sample verdicts disagree in 6 of 16 cells. The largest in-sample sentiment advantage (Heston, 2010–2012, DM $t = 12.46$) vanishes out-of-sample ($t = -0.44$), while Bates reverses from in-sample unfavourable to out-of-sample favourable in three regimes. Robust, sign-consistent sentiment gains survive in only a minority of cells. All findings are reported with Benjamini-Hochberg control of the false discovery rate.

Two further findings bound any interpretation. First, a previous-day persistence forecast attains lower surface error than every calibrated variant in all 16 cells under the vega-weighted implied-volatility metric, by a mean factor of 2.0—but the same comparison in price space yields a mean ratio of only 1.08, and a converged classical least-squares calibration of the same pricer attains comparable accuracy. The choice of error metric, not the calibration method, drives the apparent dominance of the naive benchmark. Second, an identification analysis shows that most structural parameters are only weakly recovered, and the sentiment channel does not improve identification.

Keywords: deep calibration; stochastic volatility; implied volatility surface; FinBERT; sentiment analysis; walk-forward evaluation

JEL: C45, C58, G13

1 Introduction

1.1 Motivation

Deep calibration has made parametric stochastic-volatility models tractable at speeds classical optimisation cannot approach. By learning the map from implied volatility surfaces to model parameters offline, a neural network replaces a costly inner optimisation loop with a single forward pass [Horvath et al., 2021, Bayer and Stemper, 2018]. This opens a question that was previously impractical to ask: if the calibration network can absorb an arbitrary input channel at negligible cost, can it exploit information beyond the surface itself?

Financial news sentiment is a natural candidate. A substantial literature links investor sentiment to returns and to the cross-section of asset prices [Baker and Wurgler, 2006, Tetlock, 2007]. If sentiment carries information about the risk-neutral density that the current surface has not yet fully impounded, a calibration network given both should outperform one given only the surface.

1.2 Research question

What is the effect of supplying a daily financial-news sentiment signal to the training and calibration of a neural stochastic-differential-equation option-pricing model, and does that effect survive out-of-sample evaluation?

1.3 Contributions

1. We construct a two-arm experiment in which the sentiment and baseline models are identical in architecture, hyperparameters, data and random seeds, differing only in the presence of a sentiment input channel.
2. We evaluate the arms not only on in-sample surface fit but under expanding-window walk-forward calibration, and show that the two disagree in 6 of 16 cells—including cases where the strongest in-sample effect disappears entirely.
3. We benchmark both arms against a previous-day persistence forecast and find that neither beats it in any cell under the standard vega-weighted implied-volatility metric.
4. We show that this benchmark comparison reverses in magnitude depending on whether error is measured in implied-volatility or price units, and that a converged classical least-squares calibration of the same pricer attains comparable accuracy—locating the residual error in the model class rather than in the calibration method.
5. We document and correct seven data-generation defects in the offline stage that, left in place, produce qualitatively different and misleading conclusions (Appendix A).

1.4 Hypotheses

The study was designed around two pre-registered hypotheses:

- H1 (pricing).** The sentiment arm attains lower surface pricing error than the baseline, with the largest improvements in high-volatility regimes.
- H2 (hedging).** The sentiment arm attains lower delta-hedged profit-and-loss variance than the baseline.

All other analyses are exploratory and are labelled as such.

2 Related work

Deep calibration. Neural approximation of pricing maps dates to [Hutchinson et al. \[1994\]](#), but the modern two-stage form is due to [Horvath et al. \[2021\]](#), with related contributions by [Bayer and Stemper \[2018\]](#) and [Ferguson and Green \[2018\]](#).

Stochastic volatility. We evaluate four families: [Heston \[1993\]](#), its jump extension [Bates \[1996\]](#), the one-factor forward-variance model of [Bergomi \[2005\]](#), and rough Bergomi [[Bayer et al., 2016](#)]. The last reflects evidence that volatility is rough, with Hurst exponents well below $1/2$ [[Gatheral et al., 2018](#)]. [Bakshi et al. \[1997\]](#) remains the reference point for comparative empirical performance.

Sentiment. We use FinBERT [[Araci, 2019](#)], a BERT variant [[Devlin et al., 2019](#)] fine-tuned for financial text, following the finding that domain-specific lexicons outperform general-purpose ones in finance [[Loughran and McDonald, 2011](#)]. Whether news tone carries pricing-relevant information beyond price history goes back at least to [Tetlock \[2007\]](#).

Surface dynamics. Our persistence-benchmark finding connects to work on the predictable dynamics of the implied volatility surface [[Gonçalves and Guidolin, 2006](#), [Cont and da Fonseca, 2002](#)].

3 Data

3.1 Option data

We use daily S&P 500 index option quotes partitioned into four regimes:

Regime	Character	Trading days
2010–2012	Post-crisis recovery	744
2013–2015	Low-volatility expansion	727
2016–2019	Extended calm, four years	995
2020–2022	COVID shock and aftermath	749

For each trade date we build an 8×11 grid of 8 maturity slices, $\tau \in \{0.1, 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.0\}$ years, by 11 log-moneyness levels spanning $[-0.5, +0.5]$. Days with insufficient coverage are dropped.

3.2 Forward reconstruction and normalisation

Option prices are normalised by the forward rather than by a spot proxy. Discount factors are built from FRED Treasury yields (DGS1MO–DGS3Y) interpolated to each option’s maturity, with a constant 1.9% dividend yield. Working in forward units removes the dividend yield from the pricing map, since the forward is by construction the martingale numeraire.

We replace the percentile-based global price cap used in earlier iterations with a no-arbitrage band filter that drops quotes outside [intrinsic, forward]. The band filter removes 0.28% of rows; after filtering, 100% of retained prices are invertible to implied volatility, with a median shift of -0.0017 volatility points relative to the vendor’s own implied volatilities.

3.3 News sentiment

Article-level sentiment is extracted with FinBERT from Refinitiv/LSEG Machine Readable News, filtered to U.S. equity-market relevance. The corpus spans 4 January 2010 to 31 January 2025 and comprises **227,424 articles across 3,794 trading days**, a mean of 59.9 and median of 50 articles per day (maximum 422). Articles are split into 400-character chunks, yielding **2,384,978 chunks**, an average of 10.5 per article.

FinBERT returns a probability distribution over positive, negative and neutral for each chunk. Chunk probabilities are averaged within an article, and the continuous article score is $P(\text{pos}) - P(\text{neg})$. Article scores are averaged by publication date to yield a daily sentiment index.

The news corpus extends beyond the option sample; the evaluation period ends in 2022 because the regime partition is bounded by the option data.

4 Synthetic data generation

4.1 Parameter sampling and surface construction

For each family we sample 50,000 parameter vectors uniformly from the bounds in Table 1 and price the corresponding 8×11 surface, using Fourier methods where a characteristic function is available [Carr and Madan, 1999] and Monte Carlo otherwise.

Table 1: Parameter sampling bounds. 50,000 vectors per family, uniform.

Family	Parameter	Range
Heston	κ mean-reversion speed	[0.1, 3.0]
	θ long-run variance	[0.01, 0.15]
	σ vol-of-vol	[0.1, 0.8]
	v_0 initial variance	[0.01, 0.15]
	ρ spot–variance correlation	[−0.90, −0.10]
Bates	$\kappa, \theta, \sigma, v_0, \rho$	as Heston
	λ_J jump intensity	[0, 2.0]
	μ_J mean jump size	[−0.20, 0.05]
	σ_J jump volatility	[0.01, 0.20]
Bergomi	ξ initial forward variance	[0.01, 0.16]
	ν vol-of-vol	[0.5, 4.0]
	ρ	[−0.95, −0.10]
	β	[0, 10.0]
rBergomi	ξ, ν, ρ	as Bergomi
	H Hurst exponent	[0.025, 0.50]

The spot–variance correlation is sampled strictly negative in both families, imposing the leverage effect as a structural restriction rather than leaving its sign to be recovered. The mean jump size is likewise predominantly negative, consistent with the downward-skewed jump risk priced into index options. The Hurst exponent spans [0.025, 0.50], covering the rough regime documented empirically [Gatheral et al., 2018] up to the classical Brownian endpoint. These bounds give the reference distribution for the identification analysis in Section 6.6.

Three properties of the generator warrant emphasis, because each was a defect in an earlier iteration (Appendix A):

- **Martingale correction.** Simulated terminal prices are rescaled so that $\mathbb{E}[S_T]$ matches the forward. Without it the discretised variance process induces a drift error reaching -19% at $T = 2$.
- **Full truncation.** The variance process uses full truncation [Lord et al., 2010] rather than naive clamping.
- **Log-moneyness grid.** Synthetic surfaces use the same log-moneyness grid as the market surfaces.

After correction, the fraction of synthetic surface points failing to invert falls from 16.8–37.6% to 0.22–0.70%.

4.2 Synthetic sentiment

Endogenous sentiment-driven dynamics. Rather than generating price paths and attaching sentiment post hoc, the simulation couples every path to a latent sentiment process. Sentiment Y_t follows an Ornstein-Uhlenbeck process augmented with compound Poisson jumps,

$$dY_t = \kappa_Y(\theta_Y - Y_t)dt + \sigma_Y dW_t^Y + dJ_t^Y, \quad (1)$$

with parameters sampled independently per surface. This latent process enters the variance dynamics multiplicatively,

$$dV_t = \kappa(\theta e^{\beta_V Y_t} - V_t)dt + \sigma\sqrt{V_t}dW_t^V, \quad (2)$$

and the log-price drift additively through $\beta_S Y_t$; its innovations are correlated with the price shocks. Sentiment therefore has a structural, causal effect on the simulated dynamics rather than being a label attached to an independently generated surface.

Crucially, Y_t is *not* supplied to the network. To preserve a strict analogy with the empirical setting—where FinBERT scores are an imperfect, noisy observation of an unobservable market mood—we compute return-based path statistics from the sentiment-driven trajectories and pass them through the empirical regression below to produce the synthetic scores the network sees.

The empirical sentiment map. Because sentiment is bounded in $(-1, 1)$, we model it in unbounded space: the response is $y = \text{arctanh}(s)$, mapped back through $\tanh$. The predictors are computable from any price path. After the 21-day rolling-window burn-in, $n = 3,774$ of the 3,794 available days enter the regression.

The fit explains roughly half the variation in daily sentiment and is dominated by persistence: lagged sentiment enters with coefficient 0.686 and $t = 58.6$. The clearest market linkages are contemporaneous returns ($t = -16.7$) and absolute returns ($t = -5.4$).

A limitation of this construction. The dominant predictor, lagged sentiment, has no analogue on a simulated path with no news history. The generator substitutes a path-derived proxy (a z -scored lagged return). The synthetic channel therefore reproduces the return-driven component of real sentiment but not its autoregressive persistence.

4.3 Channel representation

The baseline arm receives a single-channel 8×11 implied-volatility grid. The sentiment arm receives two channels: the surface, and a sentiment channel broadcast across strikes within each maturity slice. The two arms are otherwise identical. The sentiment channel is standardised per regime; the offline channel has pooled mean -0.0296 and standard deviation 0.0927 , and the resulting per-regime scale factors are 0.647, 0.917, 0.790 and 0.863.

Table 2: Regression of arctanh-transformed daily FinBERT sentiment on return-based path statistics. Residual s.e. 0.1122 on 3,762 d.f. $R^2 = 0.5080$, adjusted $R^2 = 0.5065$, $n = 3,774$. In sentiment space $R^2 = 0.4979$.

Variable	Estimate	Std. Error	t	$\Pr(> t)$
(Intercept)	0.06556	0.00608	10.783	$< 2 \times 10^{-16}$
y_lag1	0.68555	0.01170	58.599	$< 2 \times 10^{-16}$
ret	-3.22856	0.19308	-16.721	$< 2 \times 10^{-16}$
ret_sq	13.05354	7.07931	1.844	0.0652
abs_ret	-2.27957	0.41895	-5.441	5.29×10^{-8}
roll_mean_5	1.04134	0.60250	1.728	0.0839
roll_mean_21	-6.59953	2.54445	-2.594	0.0095
roll_vol_5	1.03807	0.55849	1.859	0.0631
roll_vol_21	4.60709	2.40796	1.913	0.0557
downside_21	-5.99732	3.11629	-1.925	0.0543
drawdown_21	0.12228	0.17800	0.687	0.4920
realized_var_21	-10.62001	15.99044	-0.664	0.5070

5 Method

5.1 Offline pre-training

A convolutional network maps the input surface to the parameter vector: 2-D convolutions with batch normalisation and ELU activations, a dense head with dropout, a sigmoid output, and a scaling layer mapping to each parameter’s admissible range. Training minimises error against the known generating parameters using Adam with a ReduceLROnPlateau schedule.

Table 3: Offline pre-training, best validation loss.

Model	Baseline (In1)	Sentiment (In2)	Improvement
Heston	0.00349	0.00261	25%
Bates	0.01213	0.01123	7%
Bergomi	0.00336	0.00322	4%
rBergomi	0.00347	0.00321	7%

The sentiment arm attains lower validation loss in every family, establishing that the channel is learnable. Whether that capability transfers to real sentiment is the question the online stage answers.

5.2 Online fine-tuning

The pre-trained weights initialise an unsupervised refinement loop against market surfaces. The network predicts parameters; a differentiable pricer prices the 8×11 grid; and the reconstruction error is backpropagated through the pricer into the network. The objective is Huber loss on the surface. Each of the 32 cells is fine-tuned independently for 750 epochs.

A note on early stopping. In the calmest regime (2013–2015), three cells initially appeared to collapse at epoch 1 under a rule terminating after 10 consecutive non-improving epochs. This was a false positive: the learning-rate scheduler does not engage until epoch 15, every cell in that regime exhibits rising validation loss over the first 10–20 epochs, and a surviving cell showed an identical early trajectory before descending to the best loss in its regime. Relaxing the rule to

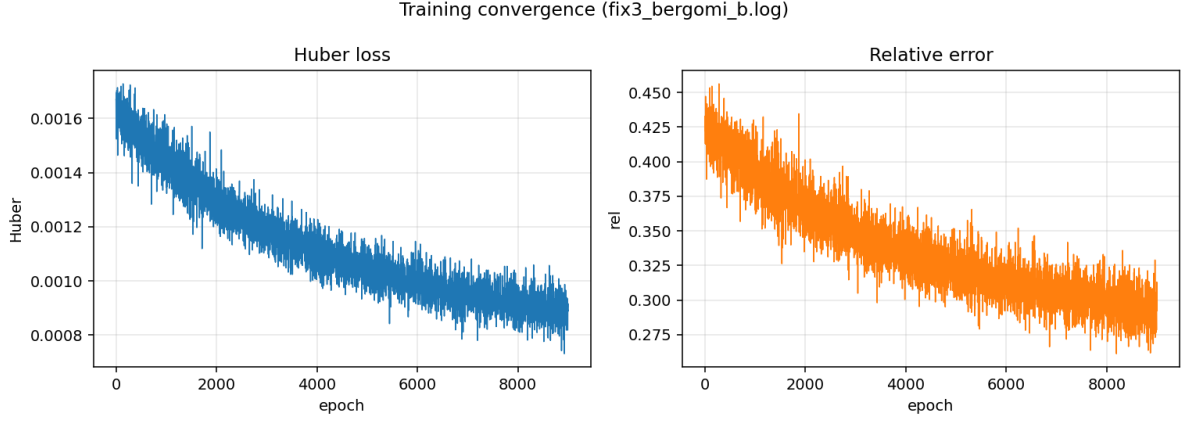


Figure 1: Training trajectory for a representative cell. Huber reconstruction loss (left) and relative error (right) across fine-tuning epochs.

30 consecutive increases, armed after epoch 20, recovered all three cells, which trained to epochs 717–744 with 12–21% improvements over initialisation.

5.3 Evaluation protocols

In-sample. All cells are evaluated on the full regime, reporting vega-weighted implied-volatility RMSE. Because the arms are separately estimated rather than nested in the estimation sense, we use two-sided Diebold-Mariano tests [Diebold and Mariano, 1995] as the primary comparison; Clark-West statistics [Clark and West, 2007] are computed but interpreted with caution.

All test statistics use a Newey-West HAC variance estimator [Newey and West, 1987] with the automatic bandwidth rule of Andrews [1991], $[4(T/100)^{2/9}]$. Daily surface loss differentials are strongly serially correlated, and unadjusted t -statistics would be materially inflated.

Walk-forward. Within each regime the sample is split into quarterly blocks. The model trains on all quarters up to Q and is evaluated on $Q + 1$, rolling forward, warm-started from the previous window. This yields 36 windows per model per arm.

Benchmarks. Both arms are compared against a previous-day persistence forecast and an at-the-money-flat surface, scored identically. Section 6.3 additionally reports the comparison in normalised price units and against a classical least-squares calibration.

Hedging. For each cell we run a rolling at-the-money delta hedge over every trading day, using 20,000 Monte Carlo paths, comparing the variance of the hedging error between arms.

Multiple testing. We report Benjamini-Hochberg [Benjamini and Hochberg, 1995] adjusted p -values controlling the false discovery rate at 5%. All twelve nominally significant out-of-sample results survive. In-sample eight of nine survive; the exception is rBergomi 2013–2015 (raw $p = 0.031$, adjusted $p = 0.055$), which we do not count among our significant findings.

6 Results

6.1 In-sample pricing

Eleven of sixteen cells show a nominally significant difference, split seven favourable and four unfavourable; eight survive FDR control. Effect sizes are uniformly small: the largest, Heston

Table 4: In-sample vega-weighted IV RMSE and Diebold-Mariano tests. Positive t favours sentiment. $\dagger$ marks cells surviving Benjamini-Hochberg correction at FDR 5%.

Model	Regime	n	RMSE (NS)	RMSE (S)	DM t	BH
Bates	2010–2012	744	0.04758	0.04774	−1.87	
Bates	2013–2015	727	0.06571	0.06554	1.64	
Bates	2016–2019	995	0.06033	0.06111	−6.89	$\dagger$
Bates	2020–2022	749	0.06799	0.06773	1.72	
Bergomi	2010–2012	744	0.05123	0.05141	−0.96	
Bergomi	2013–2015	727	0.05430	0.05416	0.34	
Bergomi	2016–2019	995	0.05331	0.05216	4.01	$\dagger$
Bergomi	2020–2022	749	0.07303	0.07889	−0.90	
Heston	2010–2012	744	0.05242	0.05004	12.46	$\dagger$
Heston	2013–2015	727	0.04786	0.04850	−4.94	$\dagger$
Heston	2016–2019	995	0.04746	0.04611	8.01	$\dagger$
Heston	2020–2022	749	0.06957	0.06706	7.57	$\dagger$
rBergomi	2010–2012	744	0.06064	0.06002	0.81	
rBergomi	2013–2015	727	0.05141	0.05044	2.16	
rBergomi	2016–2019	995	0.04836	0.04635	7.37	$\dagger$
rBergomi	2020–2022	749	0.08111	0.08464	−3.57	$\dagger$

2010–2012, is a 4.5% RMSE reduction. Heston benefits most consistently. The Bergomi family, which already imposes flexible forward-variance dynamics, shows little consistent gain.

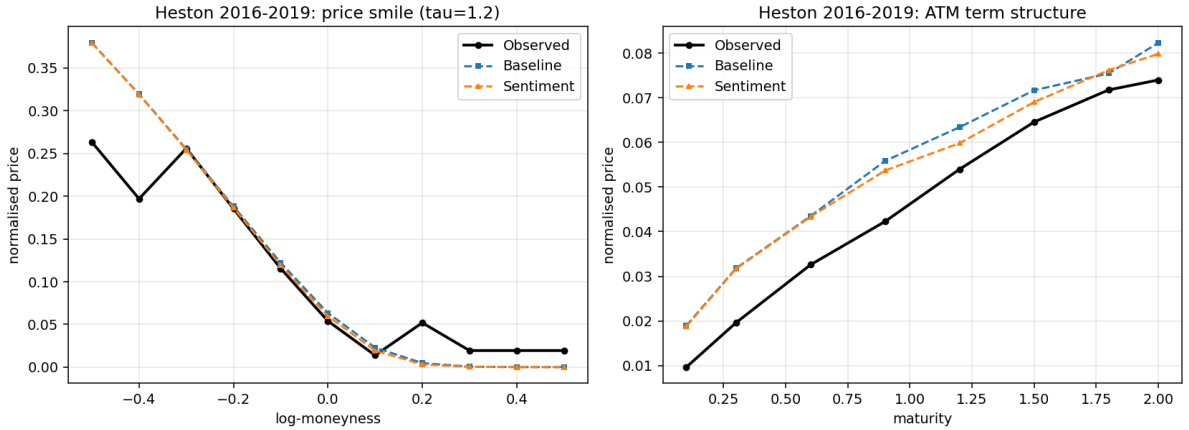


Figure 2: Price smile at a representative maturity (left) and ATM term structure (right) for Heston in 2016–2019, both arms against the observed market surface.

6.2 Out-of-sample walk-forward

Out-of-sample, seven cells favour sentiment significantly and six favour the baseline; all twelve survive FDR control. The aggregate verdict is as equivocal as in-sample—but the cell-level agreement is not. **The in-sample and out-of-sample signs disagree in six of sixteen cells:**

- **Heston 2010–2012**, the largest in-sample advantage in the study ($t = 12.46$), is indistinguishable from zero out-of-sample ($t = -0.44$).

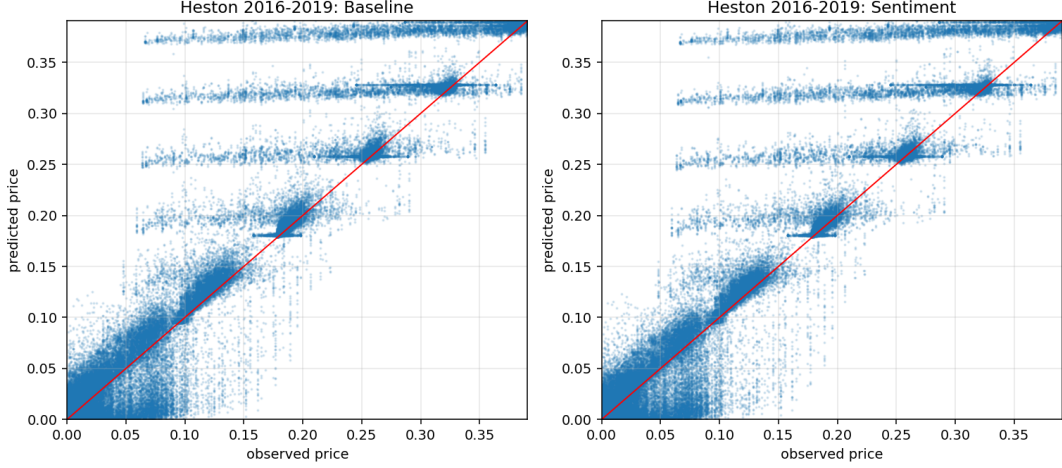


Figure 3: Predicted versus observed normalised prices, baseline (left) and sentiment (right), Heston 2016–2019.

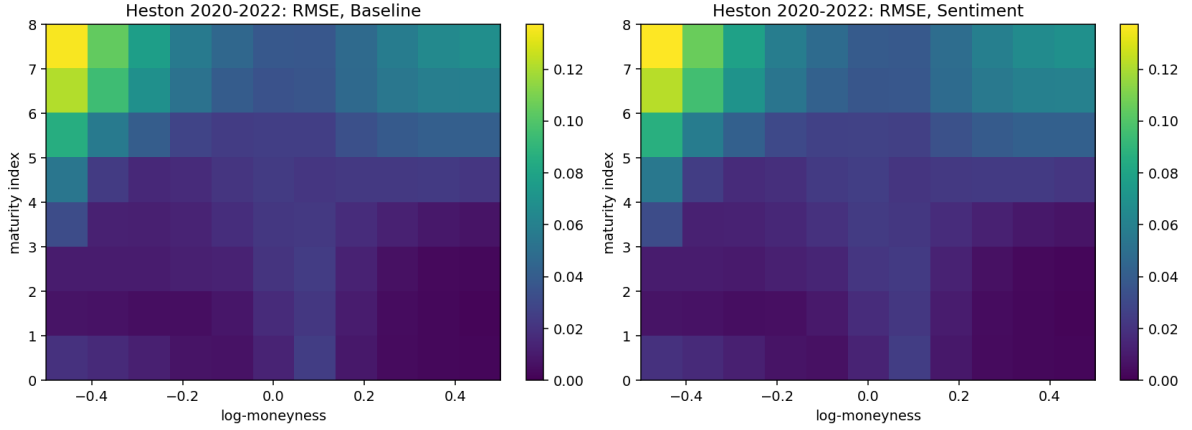


Figure 4: Per-node RMSE across the 8×11 grid, both arms, Heston 2020–2022. Errors concentrate in the deepest out-of-the-money wings and at the shortest maturities.

- **Bates reverses direction entirely.** In-sample it is the family sentiment appears to harm most; out-of-sample sentiment wins its three significant cells decisively.
- **Bergomi and rBergomi 2016–2019** flip from significantly favourable in-sample to significantly unfavourable out-of-sample.

What survives with the same sign and significance in both protocols is a minority: Heston in 2016–2019 and 2020–2022, rBergomi in 2013–2015, and Bates in 2020–2022.

We regard this as the paper’s central methodological finding. In-sample surface-fit comparison is not a reliable guide to whether an additional input channel carries transferable information; Diebold [2015] makes a related point about the interpretation of predictive-accuracy tests.

The offline/online process mismatch corrected in Appendix A (A.7) affected Bates alone, and Bates is also the family exhibiting the largest reversal. All results use the corrected generator; we note the coincidence explicitly because Bates carried an additional correction the other three families did not.

Table 5: Walk-forward out-of-sample results, with in-sample comparison. † marks cells surviving Benjamini-Hochberg correction at FDR 5%.

Model	Regime	n	RMSE (NS)	RMSE (S)	OOS t	BH	In-s. t	Agree
Bates	2010–2012	494	0.05735	0.05606	10.13	†	−1.87	no
Bates	2013–2015	478	0.10152	0.10130	1.14		1.64	yes
Bates	2016–2019	745	0.09470	0.09371	9.23	†	−6.89	no
Bates	2020–2022	496	0.06217	0.06110	9.39	†	1.72	yes
Bergomi	2010–2012	494	0.05606	0.05633	−2.15	†	−0.96	yes
Bergomi	2013–2015	478	0.11608	0.11657	−2.40	†	0.34	no
Bergomi	2016–2019	745	0.10329	0.10400	−2.92	†	4.01	no
Bergomi	2020–2022	496	0.04357	0.04349	0.92		−0.90	no
Heston	2010–2012	494	0.04629	0.04637	−0.44		12.46	no
Heston	2013–2015	478	0.08369	0.08382	−0.95		−4.94	yes
Heston	2016–2019	745	0.07014	0.06943	8.99	†	8.01	yes
Heston	2020–2022	496	0.04215	0.04150	6.40	†	7.57	yes
rBergomi	2010–2012	494	0.08136	0.08197	−3.23	†	0.81	no
rBergomi	2013–2015	478	0.05837	0.05802	2.68	†	2.16	yes
rBergomi	2016–2019	745	0.05123	0.05146	−2.30	†	7.37	no
rBergomi	2020–2022	496	0.04417	0.04470	−4.96	†	−3.57	yes

6.3 Benchmark comparison and the sensitivity of the comparison to the error metric

Under the vega-weighted implied-volatility metric, a previous-day persistence forecast attains lower error than every calibrated variant in all sixteen cells, by a mean factor of 2.0 (Table 6, left panel).

This result is metric-dependent, and materially so. Measured in normalised price RMSE on the identical predictions and days, the same comparison yields a mean ratio of **1.08**, ranging from 1.03 to 1.20.

Table 6: Benchmark comparison under two error metrics. Persistence uses the previous day’s surface and is scored identically to the models. Model ranges span the four families within each regime.

Regime	vega-weighted IV RMSE			normalised price RMSE		
	persist.	models	ratio	persist.	models	ratio
2010–2012	0.0277	0.0480–0.0607	1.73–2.19	0.0489	0.0508–0.0529	1.04–1.08
2013–2015	0.0231	0.0482–0.0656	2.09–2.84	0.0495	0.0555–0.0593	1.12–1.20
2016–2019	0.0208	0.0464–0.0604	2.23–2.91	0.0425	0.0443–0.0470	1.04–1.11
2020–2022	0.0589	0.0682–0.0818	1.16–1.39	0.0396	0.0407–0.0421	1.03–1.06

The mechanism is the vega weighting together with the price-to-IV inversion. Converting a price error into an implied-volatility error divides by vega, so in regions of small vega—the deep wings and shortest maturities, precisely where parametric models fit worst—a modest price error becomes a large IV error. Persistence is insulated because the IV surface is itself highly persistent [Cont and da Fonseca, 2002].

We report both because the divergence is the substantive point: **the choice of error metric determines whether a calibrated parametric model or a random walk appears superior.** Neither number is incorrect. A desk pricing options cares about price error; one managing vega exposure cares about IV error. Studies in this literature routinely report one without the other.

6.3.1 Is the residual error attributable to the neural calibration?

A natural objection is that the persistence gap reflects an underfitting neural calibration rather than a limitation of the model class. We test this by calibrating the same pricer to the same surfaces by classical least squares (trust-region reflective with box constraints on the Table 1 bounds, warm-started, 300 function evaluations per day) on 20 evenly-spaced days per cell.

Table 7: Neural versus classical calibration, Heston, normalised price RMSE.

Cell	classical (median)	classical (pooled)	network	persistence
Heston 2016–2019	0.0418	0.0443	0.0444	0.0425
Heston 2020–2022	0.0389	0.0438	0.0409	0.0396

Classical and neural calibration attain comparable surface fit. On typical days the classical optimiser is marginally better, but its pooled error is equal or worse, reflecting greater variability in per-day fit quality: it occasionally converges poorly, whereas the network’s single forward pass is more consistent. Neither method closes the gap to persistence in price space. We accordingly attribute the residual fitting error to the capacity of the parametric model class, not to the calibration procedure.

6.4 Delta-hedged profit and loss (H2)

Across the sixteen cells, the sentiment arm attains lower hedging-error variance in nine and higher in seven. More informative is the relationship to the pricing result: **the hedging and in-sample pricing verdicts agree in twelve of sixteen cells**. Hedging does not reveal a sentiment benefit invisible to pricing.

The largest effects appear in the COVID regime: Heston 2020–2022 shows a 12.5% variance reduction and rBergomi 2020–2022 a 15.8% reduction. The four disagreements cluster in rBergomi, the family with the weakest parameter identification.

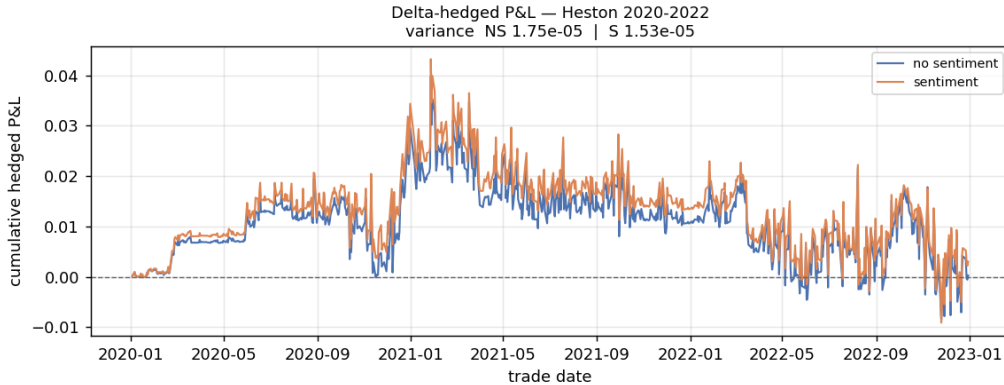


Figure 5: Delta-hedged P&L for Heston 2020–2022, the cell with the largest variance reduction attributable to sentiment (−12.5%).

H2 is not supported in general, with the qualified exception of specific high-volatility cells.

6.5 No-arbitrage diagnostics

Butterfly violations are essentially absent (0.0000–0.0016). Calendar-spread violations range from 1.4% to 19%, which at face value would concern models that are martingale by construction.

A targeted diagnostic shows these are Monte Carlo noise. Violations concentrate at the long end of the maturity axis—for a representative cell, the rate rises monotonically from 0.000 in

the $0.1 \rightarrow 0.3$ gap to 0.153 in the $1.8 \rightarrow 2.0$ gap—where the true increment in total variance is smallest and most easily flipped by simulation error. The median violation magnitude is 0.00311 against a median positive increment of 0.00812, a ratio of 0.38. Real pathology would concentrate at the short end and survive a generous tolerance; neither holds.

6.6 Parameter identification

We assess identification using the ratio of the standard deviation of observed daily parameter changes to that arising if the parameter were redrawn uniformly from its Table 1 bounds each day. A ratio near 1 indicates the network is effectively guessing.

Mean-reversion speed has ratios of 1.0–1.2 in nearly every cell: the network’s day-to-day variation in κ is as large as drawing uniformly from $[0.1, 3.0]$ afresh each day. Long-run and initial variance frequently exceed 0.9. The best-identified parameters are vol-of-vol, correlation and the jump parameters, at 0.4–0.6.

Critically, **the sentiment and baseline arms have nearly identical ratios in every row**, differing only in the third decimal. Calibrated parameter values from this pipeline should not be interpreted as estimates of physical quantities.

7 Discussion

7.1 On H1

H1 is not supported in the form stated. Sentiment does not produce a general reduction in pricing error, and the regime prediction is wrong: gains are not concentrated in high-volatility periods. The defensible statement is conditional—sentiment produces small, model-dependent effects robust to out-of-sample validation in a minority of cells.

7.2 Why offline gains do not transfer

The network demonstrably can extract information from a sentiment channel (Table 3). What differs online is the channel itself.

The synthetic channel omits the dominant term of the empirical sentiment process. As Table 2 shows, real sentiment is overwhelmingly autoregressive ($t = 58.6$), and that term cannot be reproduced on a simulated path. The offline network learns to exploit a return-driven signal and then meets an online signal whose principal component is persistence it has never seen.

Note that the offline channel is not merely redundant with the surface. Because the synthetic surfaces are generated by dynamics in which a latent sentiment process causally drives variance (Section 4.2), a genuine sentiment–volatility relationship exists for the network to learn. The failure of transfer is more plausibly about the *form* of the observed signal than about the absence of an underlying relationship.

7.3 On benchmarks and metrics

Under the IV metric a random walk dominates every calibrated variant; under a price metric the same predictions differ by 8% on average, and a converged classical optimiser performs no better than the network. What the comparison establishes is that claims about surface accuracy should be accompanied by a random-walk comparison and an explicit statement of metric.

8 Limitations

Sentiment representation. A single daily scalar. Topic-conditioned scores, intraday timing or source diversification might carry information this aggregation destroys.

The synthetic sentiment gap. The generator substitutes a return-based proxy for the lagged-sentiment term that dominates the empirical fit.

Parameter identification. Most structural parameters are weakly identified; the calibrated parameters carry little interpretive weight.

Classical benchmark configuration. Warm-started across days, 20 days per cell. It establishes comparability but is not an exhaustive benchmark; the identification diagnostics from those runs reflect the warm start and are not reported.

Residual synthetic-data floor. 0.22–0.70% of synthetic surface points remain at the inversion floor, confined to regions where vega is negligible.

Scope. Four families, four regimes, one index, one sentiment source, 750 fine-tuning epochs, 20 refit epochs per walk-forward window.

Multiple comparisons. We report Benjamini-Hochberg adjusted p -values throughout; all twelve out-of-sample and eight of nine in-sample nominally significant results survive FDR control at 5%. We report every cell and have not selected a subset post hoc.

9 Conclusion

We set out to test whether financial-news sentiment improves neural-SDE option pricing, expecting the clearest benefits in volatile markets. After rebuilding the pipeline to correct seven data-generation defects, training 32 cells, and evaluating under both in-sample and expanding-window walk-forward protocols, we find that it does not, in the general form hypothesised.

What we find instead is more specific. The sentiment channel is learnable in principle—it improves offline validation loss in all four families. Online, it produces small and heterogeneous effects surviving out-of-sample validation in only a minority of cells. In-sample and out-of-sample evaluation disagree in six of sixteen cells, including a complete reversal for one family and the disappearance of the study’s largest in-sample effect.

Every model variant is outperformed by a previous-day persistence forecast under the vega-weighted IV metric—though by only 8% on average in price units, and a converged classical calibration performs no better than the network. The apparent dominance of the naive benchmark is largely an artifact of how surface error is measured.

The methodological conclusions are the ones we would most want carried forward: that in-sample surface comparison is insufficient to establish that an auxiliary input carries transferable information; that a random-walk benchmark and an explicit statement of error metric belong alongside any reported surface RMSE; and that calibrated parameters from a deep-calibration pipeline should not be read as identified estimates without an explicit identification check.

A Corrections to the offline data-generation stage

An earlier iteration produced qualitatively different conclusions—most notably that sentiment helped most in the calmest regime. That finding did not survive correction of the following defects.

A.1 Martingale defect. Terminal simulated prices did not satisfy $\mathbb{E}[S_T] = F$. The discretised variance process under naive clamping produced a drift error reaching -19% at $T = 2$. Corrected by rescaling S_T by its sample mean at every pricing site.

A.2 Variance discretisation. Naive clamping was replaced with full truncation [Lord et al., 2010].

A.3 Quadrature weights. Gauss-Laguerre weights were constructed by a manual recursion subject to catastrophic cancellation, producing a maximum relative weight error of 6.6%. Replaced with a standard library implementation.

A.4 Sentiment channel axis error. The synthetic sentiment channel was written by indexing the Monte Carlo *path* axis rather than broadcasting across strikes. The resulting channel contained eleven unrelated path draws presented as a strike profile—noise structured to look like a signal. After correction the channel is constant across strikes within a maturity slice, with adjacent-maturity correlations of +0.74 to +0.89.

A.5 Strike grid. Synthetic surfaces were built on a linear strike grid while market surfaces used log-moneyness. Unified to log-moneyness on $[-0.5, +0.5]$.

A.6 Regression coefficients. The generator applied a superseded coefficient set whose return and volatility loadings were 5.6–7.1 times those of the current fit. Replaced with the coefficients in Table 2.

A.7 Jump compensator. The offline generator omitted the Merton jump compensator, $\lambda_J(e^{\mu_J + \sigma_J^2/2} - 1)$, from the log-price drift, while the online differentiable pricer subtracted it. The two stages therefore simulated different Bates processes: the offline surfaces carried an uncompensated drift, so the network was pre-trained on a mapping that did not correspond to the model it was later asked to calibrate. Corrected by subtracting the compensator in the generator’s drift term.

Effect of the corrections. The inversion-failure rate fell from 16.8–37.6% to 0.22–0.70%. The offline sentiment-arm advantage fell from 6–42% to 4–25%, indicating that part of the apparent offline benefit came from the inflated variance of the corrupted channel. The earlier finding that sentiment helps most in 2013–2015 disappeared entirely.

We document these in full because the corrected and uncorrected pipelines support opposite conclusions, and because several defects (A.1, A.2, A.4) would be invisible in aggregate loss curves. Training-loss diagnostics alone were not sufficient to detect any of them.

References

- Andrews, D. W. K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- Araci, D. (2019). FinBERT: financial sentiment analysis with pre-trained language models. arXiv:1908.10063.
- Baker, M. and Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *The Journal of Finance*, 61(4), 1645–1680.
- Bakshi, G., Cao, C. and Chen, Z. (1997). Empirical performance of alternative option pricing models. *The Journal of Finance*, 52(5), 2003–2049.
- Bates, D. S. (1996). Jumps and stochastic volatility: exchange rate processes implicit in Deutsche Mark options. *The Review of Financial Studies*, 9(1), 69–107.
- Bayer, C., Friz, P. and Gatheral, J. (2016). Pricing under rough volatility. *Quantitative Finance*, 16(6), 887–904.

- Bayer, C. and Stemper, B. (2018). Deep calibration of rough stochastic volatility models. arXiv:1810.03399.
- Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- Bergomi, L. (2005). Smile dynamics II. *Risk*, October 2005.
- Carr, P. and Madan, D. (1999). Option valuation using the fast Fourier transform. *Journal of Computational Finance*, 2(4), 61–73.
- Clark, T. E. and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- Cont, R. and da Fonseca, J. (2002). Dynamics of implied volatility surfaces. *Quantitative Finance*, 2(1), 45–60.
- Devlin, J., Chang, M.-W., Lee, K. and Toutanova, K. (2019). BERT: pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*, 4171–4186.
- Diebold, F. X. (2015). Comparing predictive accuracy, twenty years later. *Journal of Business & Economic Statistics*, 33(1), 1–9.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Ferguson, R. and Green, A. (2018). Deeply learning derivatives. arXiv:1809.02233.
- Gatheral, J., Jaisson, T. and Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- Gonçalves, S. and Guidolin, M. (2006). Predictable dynamics in the S&P 500 index options implied volatility surface. *The Journal of Business*, 79(3), 1591–1635.
- Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2), 327–343.
- Horvath, B., Muguruza, A. and Tomas, M. (2021). Deep learning volatility: a deep neural network perspective on pricing and calibration in (rough) volatility models. *Quantitative Finance*, 21(1), 11–27.
- Hutchinson, J. M., Lo, A. W. and Poggio, T. (1994). A nonparametric approach to pricing and hedging derivative securities via learning networks. *The Journal of Finance*, 49(3), 851–889.
- Lord, R., Koekkoek, R. and van Dijk, D. (2010). A comparison of biased simulation schemes for stochastic volatility models. *Quantitative Finance*, 10(2), 177–194.
- Loughran, T. and McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, 66(1), 35–65.
- Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Tetlock, P. C. (2007). Giving content to investor sentiment: the role of media in the stock market. *The Journal of Finance*, 62(3), 1139–1168.